

5-Mark Questions & Answers — MPCA Module 6: Pentium 4 & ARM Processor

Q1. Explain Pentium 4 NetBurst microarchitecture, Trace Cache, and Rapid Execution Engine.

- 20-stage Hyper-Pipeline.
 - 12K micro-op Trace Cache removing decode bottlenecks.
 - Rapid Execution Engine integer ALUs clocked at 2x core processor frequency.
 - Quad-pumped Front Side Bus (400-800 MHz).
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Q2. Explain Hyper-Threading Technology (HT) architecture, benefits, and hardware implementation.

- Duplicated state (GPRs, EFLAGS, CR0-CR3) vs shared execution units (ALUs, FPU, Caches).
 - 15-30% multithreaded throughput boost with minimal silicon overhead.
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Q3. Explain ARM RISC Processor architecture, Load-Store design, Barrel Shifter, and CPSR/SPSR registers.

- Strict Load-Store architecture (LDR/STR).
 - 37 total registers, Inline Barrel Shifter, conditional instruction execution, and low power efficiency.
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